

The Algebraic Bridge: FFGFT and $\mathbb{Z}_3\mathbb{C}$

Frobenius Automorphism, Charge Classification and
Five Structural Questions between $T^4/\mathbb{Z}_3$ and $\text{GF}(9)$

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Abstract

This document investigates five algebraic structural questions between FFGFT geometry ($T^4/\mathbb{Z}_3$, Docs. 257/321) and Matzke's GALG framework ($G(6, \mathbb{Z}_3\mathbb{C})$). Three main results **[B]**: (1) The FFGFT sector pairing $k \leftrightarrow -k$ and Matzke's $V_s = V^\dagger$ both realise the Frobenius automorphism $x \mapsto x^3$ in $\text{GF}(9)$. (2) The canonical ring homomorphism $\varphi : N \mapsto N \bmod 3$ connects Furey's number operator with the FFGFT winding quantum. (3) The injection $\iota : Q \mapsto (3Q \bmod 3, [Q])$ shows that $N \bmod 3$ fully classifies the charge $Q = N/3$.

Verification scripts: `pruef_324_z3c_trine_exakt.py`,
`pruef_325_gf9_ffgft_bruecke.py`.

.1 Motivation: Characteristic 3 and FFGFT

In characteristic 3, $(1 + N)^3 = 1$ for nilpotent N with $N^2 = 0$ holds *algebraically exactly* — because $\binom{3}{k} \equiv 0 \pmod{3}$ for $0 < k < 3$. The machine-epsilon result $|T_k^3 - \mathbf{1}| < 6.5 \times 10^{-16}$ from Doc. 324 is therefore not an approximation but a consequence of an algebraic theorem **[B]**.

.2 Five structural questions — updated status

Question 1: Number operator N and FFGFT winding quantum Δ_w

Previously open [S]; now partially closed.

Furey's number operator $N = \sum_j q_j^\dagger q_j$ has spectrum $\{0, 1, 1, 1, 2, 2, 2, 3\}$ on the eight basis states of one generation. Reduction mod 3 gives:

$$N \bmod 3 = \{0, 1, 1, 1, 2, 2, 2, 0\} \quad (1)$$

with particle classification:

$N \bmod 3 = 0$: colour-neutral (neutrino, positron)

$N \bmod 3 = 1$: anti-down quarks (3 colours)

$N \bmod 3 = 2$: up quarks (3 colours)

This is *exactly* the $\mathbb{Z}_3$ class decomposition of the FFGFT sectors $\Delta_w \in \{0, 1, 2\}$ on $T^4/\mathbb{Z}_3$.

Ring homomorphism [B]:

$$\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}, \quad \varphi(N) = N \bmod 3 = \Delta_w \quad (2)$$

is the canonical quotient; $\varphi(N + M) = \varphi(N) + \varphi(M) \bmod 3$ is a proved algebraic property.

Status: [B] (homomorphism), [K] (physical assignment)

$\varphi(N) = N \bmod 3$ is a ring homomorphism **[B]**. The physical identification $\varphi(N) = \Delta_w$ (Furey's number operator $\leftrightarrow$ FFGFT winding quantum) classifies the same threefold particle spectrum decomposition; the deeper algebraic reason for this agreement is **[K]**.

Question 2: Charge formula $Q = N/3$ and $\mathbb{Z}_3$ classification

Previously open [S]; now partially closed.

$Q = N/3 \in \mathbb{Q}$ cannot be embedded into $\text{GF}(3)$ (3 is a zero divisor in $\mathbb{Z}/9\mathbb{Z}$, no inverse). However an injection exists:

$$\iota : \{0, \frac{1}{3}, \frac{2}{3}, 1\} \hookrightarrow \mathbb{Z}/3\mathbb{Z} \times \{0, 1\}, \quad \iota(Q) = (3Q \bmod 3, [Q]) \quad (3)$$

with values:

Q	$\iota(Q)$	Particle
0	(0, 0)	Neutrino
1/3	(1, 0)	Anti-Down
2/3	(2, 0)	Up
1	(0, 1)	Positron

The $\mathbb{Z}/3\mathbb{Z}$ component of $\iota(Q)$ is exactly $N \bmod 3$.

Status: [B] (classification), [K] (Q as Q number)

$N \bmod 3$ fully classifies $Q = N/3$ **[B]**. Q itself (as rational number $1/3, 2/3$) requires Q and is not a GF(3) object **[K]**.

Question 3: Sector pairing $k \leftrightarrow -k$ and Frobenius automorphism

Main result; was [S]; now [B].

The Frobenius automorphism in GF(9):

$$\phi : x \mapsto x^3, \quad \phi(a + bi) = a - bi \quad [\text{field conjugation}] \quad (4)$$

operates simultaneously on three levels:

- **GF(9) field:** $a + bi \mapsto a - bi$ (conjugation)
- **FFGFT sector:** $k \mapsto -k \bmod 3$ ($\mathbb{Z}_3$ inversion)
- **Matzke vacuum:** $V \mapsto V_s = V^\dagger$ (matrix adjunction)
- `pruef_326_gf9_orbit_masse_weinberg.py`: Fixed points **[B]**, norm hierarchy **[B]**, Weinberg topology **[K]**

Numerically verified: $x^3 = \bar{x}$ for all 8 non-zero GF(9) elements **[B]**.

Status [B]

All three operations are algebraically isomorphic realisations of the Frobenius automorphism $x \mapsto x^3$.

Question 4: Casimir singularity and ξ in GF(3)

$\xi = (4/3)/10^4$: Casimir denominator $3 \equiv 0$ in GF(3) is singular. $\xi \bmod 3 = 1$ — ξ lies in residue class 1. The singularity marks the transition $\text{GF}(3) \rightarrow \mathbb{R}$ **[K]**.

Question 5: $\sqrt{2}$ in GF(9) and n_{thresh}

$i^2 = -1 \equiv 2 \pmod{3}$, so $\sqrt{2} = i \in \text{GF}(9)$ **[B]**. $n_{\text{thresh}} = 2\pi/(\sqrt{2} \ln 2) \approx 6.41$ (R93) thus has an algebraically exact GF(9) kernel.

.3 Three further structural results

Frobenius fixed points = FFGFT sectors (Question 3, extended)

The complete orbit structure under $\phi(x) = x^3$ in GF(9):

- **Fixed points** ($x^3 = x$): $\{0, 1, 2\} = \text{GF}(3)$ — exactly the FFGFT sectors $\Delta_w \in \{0, 1, 2\}$ **[B]**
- **Non-fixed points:** 6 elements in 3 conjugate 2-cycles **[B]**: $(i \leftrightarrow 2i)$, $(1+i \leftrightarrow 1+2i)$, $(2+i \leftrightarrow 2+2i)$ — exactly the FFGFT pairs $(k, -k)$ for $k \neq 0$.

Stable states under Frobenius are the sectors; unstable states are the conjugate pairs.

Norm hierarchy $\mathbf{GF(3)} \rightarrow \mathbf{GF(9)} \rightarrow \mathbb{R}(\xi)$ for masses

In $\mathbf{GF(3)}$, $1^2 \equiv 2^2 \equiv 1 \pmod{3}$ — muon and tau are *algebraically degenerate* as squares **[B]**. The $\mathbf{GF(3)}$ structure distinguishes only:

$$n^2 \in \{0, 1\} \pmod{3} : \quad 0 \leftrightarrow \text{massless}, \quad 1 \leftrightarrow \text{massive (generations 2+3 degenerate)} \quad (5)$$

In $\mathbf{GF(9)}$, the norm $\|x\| = (a^2 + b^2) \pmod{3}$ gives two classes **[B]**:

$$\begin{aligned} \|x\| = 1 & : \{i, 2i, 1, 2\} \quad \text{— purely real/imaginary (light)} \\ \|x\| = 2 & : \{1+i, 1+2i, 2+i, 2+2i\} \quad \text{— mixed (heavy)} \end{aligned}$$

The full mass hierarchy requires ξ **[K]**: ξ breaks the $\mathbf{GF(3)}$ degeneracy and gives $m_\mu/m_e \approx 207$, $m_\tau/m_\mu \approx 17$ as continuous numbers.

Status: **[B]** (degeneracy), **[K]** (mass values)

Hierarchy: $\mathbf{GF(3)}$ gives whether massive/massless; $\mathbf{GF(9)}$ gives light/heavy; $\xi \in \mathbb{R}$ gives exact mass ratios. The singularity $C_2 = 4/3$ in $\mathbf{GF(3)}$ (Question 4) marks exactly the transition $\mathbf{GF(3)} \rightarrow \mathbb{R}$.

Weinberg angle: $\mathbf{GF(9)}$ gives topology, ξ gives value

$\text{Gal}(\mathbf{GF(9)}/\mathbf{GF(3)}) = \mathbb{Z}_2$ explains *why* there are two gauge groups $\text{SU}(2)_L$ and $\text{U}(1)_Y$ — a degree-2 extension gives exactly two components **[K]**.

The best $\mathbf{GF(9)}$ candidate for $\sin^2 \theta_W$ is $|\mathbf{GF(3)}^*|/|\mathbf{GF(9)}^*| = 2/8 = 0.250$, compared to the experimental value 0.231 (8% deviation). The exact value is reached by ξ corrections (Doc. 323, **[K]**).

Status **[K]**

$\mathbf{GF(9)}$ provides the algebraic justification for two gauge groups (degree-2 Galois extension) and a tree-level approximation ≈ 0.25 . The numerical value $\sin^2 \theta_W = 0.231$ comes from ξ .

.4 Status overview

Verification scripts

- `pruef_324_z3c_trine_exakt.py`: Algebraic proof $T_k^3 = \mathbf{1}$ in char. 3 **[B]**
- `pruef_325_gf9_ffgft_bruecke.py`: All five questions; Frobenius **[B]**; ring homo **[B]**; injection ι **[B]**

Table 1: Algebraic bridge questions FPGFT $\leftrightarrow$ GF(9) — updated status

#	Question	Result	Status
1a	Ring homo $\varphi(N) = N \bmod 3$	canonical, proved	[B]
1b	$\varphi(N) = \Delta w$ physical	same threefold split	[K]
2a	$N \bmod 3$ classifies Q	injection ι proved	[B]
2b	$Q = N/3$ as GF(3) object	impossible ($3 \equiv 0$)	[K]
3	$k \leftrightarrow -k$ vs. $V_s = V^\dagger$	Frobenius $x \mapsto x^3$	[B]
4	$C_2 = 4/3$ and ξ in GF(3)	singularity = GF(3) $\rightarrow \mathbb{R}$	[K]
5	$\sqrt{2}$ in GF(9), n_{thresh}	$i^2 = 2$, kernel exact	[B]
P3b	Fixed pts. = sectors, $(k, -k)$ pairs	orbit structure	[B]
P4b	$1^2 = 2^2 = 1$ in GF(3)	Gen. 2+3 degenerate	[B]
P4c	Norm $\in \{1, 2\}$ in GF(9)	light/heavy	[B]
P5b	$ \text{GF}(3)^* / \text{GF}(9)^* = 1/4$	tree-level $\sin^2 \theta_W$	[K]